Group II, Row IV: fragmentation function and DGLAP

1 Your equation

Your ξ form is correct. One change is needed, in blue: the measure is $\int_{x,x',y,y'}$ in both terms, not $\int_{x,y,y'}$. The charge x' of the conjugate amplitude is integrated, as at LO.

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4 d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^+\!+\Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,x',y,y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{1-\xi}\Big]\frac{1}{\xi^2}\\
&\quad\times\Big[\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\Big]\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x)\\
&\quad+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^+\!+\Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,x',y,y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{1}{\xi^2}\Bigg[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\,d_\perp\,\xi(1-\xi)\\
&\quad+\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[\frac{1-\xi}{\xi}+\frac{\xi}{1-\xi}\Big]\Bigg]\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x). \qquad (1.1)\end{aligned}$$

It is Eq. (1.5) of `RowIV_dperp.pdf`, term by term (`sympy`: difference 0):

- the limits are reversed, $\int_{k^+/(k^+\!+\Lambda)}^{k^+/\vee} = -\int_{k^+/\vee}^{k^+/(k^+\!+\Lambda)}$, which removes the overall minus sign of the E form of (1.5);
- $\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\log\frac{(y'-y)^2\mu^2}{4}=\frac{E}{2}$, and $\frac{g^4 d_\perp}{4\pi^3}\cdot\frac{1}{2\pi}\cdot\frac{E}{2}=\frac{g^4}{16\pi^4}\,d_\perp E$;
- in the second term, $\frac{g^4}{4\pi^3}\cdot\frac{1}{2\pi}=\frac{g^4}{16\pi^4}\cdot 2$, matching the $2\big(\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\big)$ and $2d_\perp\xi(1-\xi)$ of (1.5);
- the $\frac{1}{\xi^2}$ and the phase $e^{-ik\cdot(y'-y)/\xi}$ are right.

Notation used below.

- $\Delta\equiv y'-y$, $\hat\Delta\equiv\Delta/|\Delta|$, $X\equiv\frac{x-y}{(x-y)^2}$, $X'\equiv\frac{x'-y'}{(x'-y')^2}$, and
$$\mathcal{O}\equiv\big[U^{ce'}(y')-U^{ce'}(x')\big]\rho^{e'}(x')\,\big[U^{ce}(y)-U^{ce}(x)\big]\rho^e(x), \qquad (1.2)$$
which is your colour factor, since $U^{bc}=U^{cb}$.
- $\xi_0\equiv\frac{k^+}{\vee}$ and $\xi_1\equiv\frac{k^+}{k^+\!+\Lambda}$. The light-cone variables are $\delta Y=\log\frac{\vee}{\Lambda}$ and $\delta\equiv\frac{\Lambda}{k^+}$.
- The pole is written as $\frac{1}{\bar\epsilon}\equiv-\frac{2}{\epsilon}$, i.e. $\frac{1}{\bar\epsilon}=-\frac{1}{\epsilon_{\rm std}}$ in the usual $D=4-2\epsilon_{\rm std}$ notation. Then $\frac{E}{2}=-\frac{\bar E}{2}$, with
$$\bar E\equiv\frac{1}{\bar\epsilon}-\ell, \qquad \ell\equiv\log\frac{\Delta^2\mu^2_{\overline{MS}}e^{2\gamma_E}}{4}=\log\frac{\Delta^2\mu^2_{\overline{MS}}}{c_0^2}, \qquad c_0\equiv 2e^{-\gamma_E}. \qquad (1.3)$$
The pole is collinear (infrared): $\epsilon<0$, so $\frac{1}{\bar\epsilon}>0$.
- The real part of the gluon splitting function is
$$P^{(r)}_{gg}(\xi)\equiv 2N_c\Big[\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big]=2N_c\hat\Phi(\xi). \qquad (1.4)$$

2 Row IV as a convolution with LO

LO and two weighted copies of it. The LO spectrum is

$$\frac{d^3N_{\rm LO}}{d^3k}(k^+,\mathbf{k}) = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,x',y,y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O}. \tag{2.1}$$

Insert a weight into the integrand of Eq. (2.1), with the same prefactor:

$$\mathcal{L}(k^+,\mathbf{k}) \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,x',y,y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\frac{d_\perp\bar{E}-2}{2}\,\mathcal{O}, \tag{2.2}$$

$$\mathcal{L}_D(k^+,\mathbf{k}) \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,x',y,y'} e^{-ik\cdot(y'-y)}\Big[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} - d_\perp\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\Big]\mathcal{O}. \tag{2.3}$$

$\bar{E}$ depends on $|y'-y|$, so $\mathcal{L}$ is not a multiple of LO. Its pole part is: $\mathcal{L}\big|_{\rm pole} = \frac{d_\perp}{2}\frac{1}{\bar{\epsilon}}\frac{d^3N_{\rm LO}}{d^3k}$.

The conversion. At the argument $(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi})$, Eq. (2.1) has prefactor $\frac{g^2\xi}{2\pi^2k^+}$ and phase $e^{-ik\cdot\Delta/\xi}$. Hence, for any weight W,

$$\int_{x,x',y,y'} e^{-ik\cdot\Delta/\xi}\,X\cdot X'\,W\,\mathcal{O} = (2\pi)^3\,\frac{2\pi^2k^+}{g^2\xi}\times\Big[\text{Eq. (2.1) with weight } W\Big]\Big(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi}\Big), \tag{2.4}$$

and the prefactor of Row IV becomes

$$\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\frac{1}{\xi^2}\times(2\pi)^3\frac{2\pi^2k^+}{g^2\xi} = \frac{\alpha_s}{2\pi}\,\frac{N_c}{\xi^3} \qquad (\alpha_s=\tfrac{g^2}{4\pi};\ \texttt{sympy}: \tag{2.5}$$

0).

Result. With $N_c\,d_\perp\hat{\Phi}\,\bar{E} - 2N_c\big(\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\big) = P_{gg}^{(r)}\frac{d_\perp\bar{E}-2}{2} + 2N_c\xi(1-\xi)$, Row IV is, exactly and for any $d_\perp$,

$$\boxed{\frac{d^3N_{\rm IV}}{d^3k} = \frac{\alpha_s}{2\pi}\int_{\xi_0}^{\xi_1}\frac{d\xi}{\xi^3}\Big\{P_{gg}^{(r)}(\xi)\,\mathcal{L}\Big(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi}\Big) + 2N_c\,\xi(1-\xi)\,\mathcal{L}_D\Big(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi}\Big)\Big\}} \tag{2.6}$$

(`sympy`: equal to the compact form of `RowIV_compact.pdf`, difference 0).

- **The pole** is in $\mathcal{L}$ only:

$$\frac{d^3N_{\rm IV}}{d^3k}\bigg|_{\rm pole} = \frac{\alpha_s}{2\pi}\,\frac{d_\perp}{2}\,\frac{1}{\bar{\epsilon}}\int_{\xi_0}^{\xi_1}\frac{d\xi}{\xi^3}\,P_{gg}^{(r)}(\xi)\,\frac{d^3N_{\rm LO}}{d^3k}\Big(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi}\Big). \tag{2.7}$$

- $\mathcal{L}_D$ **has no pole.** Its tensor $\delta^{ij}-d_\perp\hat{\Delta}^i\hat{\Delta}^j$ is traceless in $d_\perp$ dimensions, so it vanishes when averaged over the direction of Δ. It is the linearly polarized part of the $g\to gg$ splitting; it is finite and stays in the coefficient function.
- **The measure** $d\xi/\xi^3$ is the Jacobian of $\frac{d^3N}{d^3k} = \frac{dN}{dk^+d^2\mathbf{k}}$: the parent has $(k^+,\mathbf{k})/\xi$, so $d^3k_{\rm parent} = d^3k/\xi^3$. For $\frac{dN}{d\eta\,d^2\mathbf{k}} = k^+\frac{d^3N}{d^3k}$ the same equation has $d\xi/\xi^2$.
- **The lower limit** $\xi_0 = k^+/\vee$ is the condition that the parent momentum k^+/ξ stays below $\vee$.

The two choices of $d_\perp$.

- $d_\perp = 2-\epsilon$ (as in Group I, Rows II and V). Then $\frac{d_\perp\bar{E}-2}{2} = \frac{1}{\bar{\epsilon}}-\ell+O(\epsilon)$ (`sympy`: 0). The "+1" from $\frac{d_\perp}{2}\frac{1}{\bar{\epsilon}} = \frac{1}{\bar{\epsilon}}+1$ cancels the -1. So

$$\mathcal{L} = \frac{1}{\bar{\epsilon}}\frac{d^3N_{\rm LO}}{d^3k} - \frac{d^3N_\ell}{d^3k}, \qquad \mathcal{L}_D = \frac{d^3N_{\rm LO}}{d^3k} - 2\,\frac{d^3N_D}{d^3k}, \tag{2.8}$$

  where $\frac{d^3N_\ell}{d^3k}$ is Eq. (2.1) with weight ℓ and $\frac{d^3N_D}{d^3k}$ is Eq. (2.1) with $X\cdot X'\to(X\cdot\hat{\Delta})(X'\cdot\hat{\Delta})$. The residue of the pole is $P_{gg}^{(r)}$ exactly.

- $d_\perp = 2$. Then $\frac{d_\perp\bar{E}-2}{2} = \frac{1}{\bar{\epsilon}}-\ell-1$. The residue is again $P_{gg}^{(r)}$. The finite part differs by $-\frac{\alpha_s}{2\pi}\int\frac{d\xi}{\xi^3}P_{gg}^{(r)}\,\frac{d^3N_{\rm LO}}{d^3k}$, which is a change of factorization scheme.

The DGLAP kernel is the residue, so it is the same in both choices. The finite parts are not, so every row has to be evaluated with the same $d_\perp$.

3 The endpoint $\xi \to 1$ and the plus prescription (exact limits)

The upper limit $\xi_1 = \frac{k^+}{k^+ + \Lambda} < 1$ regulates the soft gluon, $p^+ \to \Lambda$. To compare with DGLAP, use the standard plus distribution on $[0,1]$, cut at ξ_0:

$$\int_{\xi_0}^1 d\xi \Big[\frac{1}{1-\xi}\Big]_+ f(\xi) \equiv \int_{\xi_0}^1 d\xi \, \frac{f(\xi) - f(1)}{1-\xi} + f(1)\log(1-\xi_0), \tag{3.1}$$

and $P^{(r)}_{gg,+}(\xi) \equiv 2N_c\big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\big]$, with $\frac{\xi}{(1-\xi)_+} \equiv \xi\big[\frac{1}{1-\xi}\big]_+$. Let $f(\xi)$ be any function regular at $\xi = 1$. Exactly,

$$\int_{\xi_0}^{\xi_1} d\xi \, P^{(r)}_{gg}(\xi) f(\xi) = \int_{\xi_0}^1 d\xi \, P^{(r)}_{gg,+}(\xi) f(\xi) + 2N_c \log\frac{1}{1-\xi_1}\, f(1) - \int_{\xi_1}^1 d\xi \Big[P^{(r)}_{gg}(\xi) f(\xi) - \frac{2N_c f(1)}{1-\xi}\Big] \tag{3.2}$$

(numerically, rel. 1×10^{-13}). With $f(\xi) = \xi^{-3}\frac{d^3N_{\rm LO}}{d^3k}(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi})$ we have $f(1) = \frac{d^3N_{\rm LO}}{d^3k}(k^+, \mathbf{k})$, and

$$\log\frac{1}{1-\xi_1} = \log\frac{k^+ + \Lambda}{\Lambda} = \log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+} + \log\Big(1 + \frac{\Lambda}{k^+}\Big). \tag{3.3}$$

So the pole (2.7) of Row IV is, exactly,

$$\begin{aligned}\frac{d^3N_{\rm IV}}{d^3k}\bigg|_{\rm pole} = \frac{\alpha_s}{2\pi}\frac{d_\perp}{2}\frac{1}{\bar\epsilon}\Big\{ &\int_{\xi_0}^1 \frac{d\xi}{\xi^3} P^{(r)}_{gg,+}(\xi) \frac{d^3N_{\rm LO}}{d^3k}\Big(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi}\Big) \\ &+ 2N_c\Big[\log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+} + \log\Big(1 + \frac{\Lambda}{k^+}\Big)\Big]\frac{d^3N_{\rm LO}}{d^3k}(k^+, \mathbf{k}) - R_\Lambda\Big\},\end{aligned} \tag{3.4}$$

with

$$R_\Lambda \equiv \int_{\xi_1}^1 d\xi \Big[P^{(r)}_{gg}(\xi)\,\xi^{-3}\frac{d^3N_{\rm LO}}{d^3k}\Big(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi}\Big) - \frac{2N_c}{1-\xi}\frac{d^3N_{\rm LO}}{d^3k}(k^+, \mathbf{k})\Big] = -2N_c\frac{\Lambda}{k^+}\Big[f(1) + f'(1)\Big] + O\Big(\frac{\Lambda^2}{k^{+2}}\Big). \tag{3.5}$$

R_Λ is regular. The expansion was checked numerically: $R_\Lambda/\delta = -1.97000,\ -1.97021,\ -1.97025$ for $\delta = 10^{-2}, 10^{-3}, 10^{-4}$, vs. -1.97025. It is not of the form (number)×LO.

- The first line of Eq. (3.4) is the real part of $P_{gg} \otimes \mathrm{LO}$.
- The second line is a $\delta(1-\xi)$ term: the soft logarithm $2N_c \log\frac{k^+ + \Lambda}{\Lambda}$. It cancels against the virtual rows of Group I (Sec. 6).
- The full Row IV, Eq. (2.6), is rewritten the same way with $f(\xi) = \xi^{-3}\mathcal{L}(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi})$. The $\mathcal{L}_D$ term is regular at $\xi = 1$ and needs no plus prescription.

4 The $\overline{\rm MS}$ fragmentation function

Factorization. A hadron (or any final state) h with momentum $(k^+, \mathbf{k})$ is produced from a gluon with $(k^+, \mathbf{k})/\xi$:

$$\frac{d^3N_h}{d^3k}(k^+, \mathbf{k}) = \int_{\xi_0}^1 \frac{d\xi}{\xi^3}\, D^{(0)}_{h/g}(\xi)\, \frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi}\Big), \tag{4.1}$$

where $D^{(0)}_{h/g}$ is the bare fragmentation function and $\frac{d^3N_g}{d^3k} = \frac{d^3N_{\rm LO}}{d^3k} + \frac{d^3N_{\rm NLO}}{d^3k}$ is the gluon spectrum computed in these notes.

The pole of the gluon spectrum. Sec. 6 adds the poles of Row IV, of the virtual rows of Group I, and of the charge renormalization of LO. Up to terms of order $\frac{\Lambda}{k^+}$ and $\frac{k^+}{\vee}$, the sum is

$$\frac{d^3N_g}{d^3k}\bigg|_{\rm pole} = \frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}\int_{\xi_0}^1 \frac{d\xi}{\xi^3}\, P_{gg}(\xi)\, \frac{d^3N_{\rm LO}}{d^3k}\Big(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi}\Big), \tag{4.2}$$

$$P_{gg}(\xi) = 2N_c\Big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\Big] + \frac{11N_c}{6}\,\delta(1-\xi) \qquad (n_f = 0). \tag{4.3}$$

This is the full LO gluon splitting function. The coefficient $\frac{11N_c}{6} = \frac{\beta_0}{2}$ of $\delta(1-\xi)$ is also fixed independently by momentum conservation: $\int_0^1 d\xi\, \xi\, P^{(r)}_{gg,+} = -\frac{11N_c}{6}$ (`sympy`), so $\int_0^1 d\xi\, \xi P_{gg} = 0$ requires $\frac{11N_c}{6}\delta(1-\xi)$.

The counterterm. Define

$$\boxed{\Gamma_{gg}(\xi) \equiv \delta(1-\xi) - \frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon} P_{gg}(\xi) = \delta(1-\xi) + \frac{\alpha_s}{2\pi}\frac{2}{\epsilon} P_{gg}(\xi)} \tag{4.4}$$

(the second form is in your ϵ, $d_\perp = 2-\epsilon$; $\overline{\rm MS}$ subtracts only $\frac{2}{\epsilon}$, since $\mu_{\overline{MS}}$ already contains $\frac{e^{\gamma_E}}{4\pi}$). The renormalized fragmentation function $D_{h/g}(\xi,\mu)$ is defined by

$$D^{(0)}_{h/g}(\xi) = \int_\xi^1 \frac{dz}{z}\,\Gamma_{gg}(z)\, D_{h/g}\Big(\frac{\xi}{z},\mu\Big) = D_{h/g}(\xi,\mu) - \frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}\int_\xi^1 \frac{dz}{z} P_{gg}(z)\, D_{h/g}\Big(\frac{\xi}{z},\mu\Big) + O(\alpha_s^2). \tag{4.5}$$

Insert Eq. (4.5) into Eq. (4.1) and use the Jacobian identity (with $\xi = \zeta z$)

$$\int_{\xi_0}^1 \frac{d\xi}{\xi^3}\Big[\int_\xi^1 \frac{dz}{z} A(z) D\Big(\frac{\xi}{z}\Big)\Big] F\Big(\frac{k}{\xi}\Big) = \int_{\xi_0}^1 \frac{d\zeta}{\zeta^3} D(\zeta) \int_{\xi_0/\zeta}^1 \frac{dz}{z^3} A(z)\, F\Big(\frac{k}{\zeta z}\Big) \tag{4.6}$$

(numerically, rel. 4×10^{-14}). Then

$$\frac{d^3N_h}{d^3k} = \int_{\xi_0}^1 \frac{d\xi}{\xi^3}\, D_{h/g}(\xi,\mu)\, \frac{d^3N_g^{\overline{MS}}}{d^3k}\Big(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi}; \mu\Big), \qquad \frac{d^3N_g^{\overline{MS}}}{d^3k}(k;\mu) \equiv \int \frac{dz}{z^3}\,\Gamma_{gg}(z)\, \frac{d^3N_g}{d^3k}\Big(\frac{k}{z}\Big). \tag{4.7}$$

The counterterm $-\frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon} P_{gg} \otimes \mathrm{LO}$ cancels Eq. (4.2), so $\frac{d^3N_g^{\overline{MS}}}{d^3k}$ is finite.

Row IV after the subtraction ($d_\perp = 2-\epsilon$). With Eq. (2.8), the part of Row IV that stays in $\frac{d^3N_g^{\overline{MS}}}{d^3k}$ is Eq. (2.6) without its $\frac{1}{\bar\epsilon}$:

$$\frac{d^3N_{\rm IV}^{\overline{MS}}}{d^3k} = \frac{\alpha_s}{2\pi}\int_{\xi_0}^{\xi_1}\frac{d\xi}{\xi^3}\Big\{ - P^{(r)}_{gg}(\xi)\,\frac{d^3N_\ell}{d^3k}\Big(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi}\Big) + 2N_c\,\xi(1-\xi)\Big[\frac{d^3N_{\rm LO}}{d^3k} - 2\frac{d^3N_D}{d^3k}\Big]\Big(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi}\Big)\Big\}. \tag{4.8}$$

The logarithm $\ell = \log\frac{(y'-y)^2\mu^2}{c_0^2}$ vanishes at $\mu = c_0/|y'-y|$, the natural scale of the fragmentation logarithm. With $|y'-y| \sim 1/|\mathbf{k}|$, this is $\mu \sim |\mathbf{k}|$.

5 DGLAP

From the μ independence of the cross section. $\frac{d^3N_h}{d^3k}$ does not depend on μ.

- In $\frac{d^3N_g^{\overline{MS}}}{d^3k}$ every pole came with its logarithm, $\bar E = \frac{1}{\bar\epsilon} - \log\frac{L_\perp^2\mu^2}{c_0^2}$, with $L_\perp = y'-y$ in Row IV, $x-w$ in Row II and $y-z$ in Row V 2b. After the subtraction, $\frac{\partial}{\partial\log\mu^2}$ gives -1 times the residue. Summed over the rows, the residues are $P^{(r)}_{gg,+}$ plus the $\delta(1-\xi)$ terms of Sec. 6 without the charge renormalization. The soft logarithms and the $\frac{11}{3}$ cancel there.
- The LO term carries $\alpha_s(\mu)$, with $\frac{d\alpha_s}{d\log\mu^2} = -\frac{\beta_0}{4\pi}\alpha_s^2$. This gives $-\frac{\alpha_s}{2\pi}\frac{\beta_0}{2}\mathrm{LO}$.

Together,

$$\frac{\partial}{\partial\log\mu^2}\frac{d^3N_g^{\overline{MS}}}{d^3k}(k^+,\mathbf{k};\mu) = -\frac{\alpha_s}{2\pi}\int_{\xi_0}^1\frac{dz}{z^3}P_{gg}(z)\,\frac{d^3N_{\rm LO}}{d^3k}\Big(\frac{k^+}{z},\frac{\mathbf{k}}{z}\Big) + O(\alpha_s^2). \tag{5.1}$$

By Eq. (4.6), μ independence of $\frac{d^3N_h}{d^3k}$ requires

$$\boxed{\mu^2\frac{\partial}{\partial\mu^2}D_{h/g}(\xi,\mu) = \frac{\alpha_s(\mu)}{2\pi}\int_\xi^1\frac{dz}{z}\,P_{gg}(z)\,D_{h/g}\Big(\frac{\xi}{z},\mu\Big)} \tag{5.2}$$

with P_{gg} of Eq. (4.3). This is the time-like DGLAP equation for gluon fragmentation (pure glue, $n_f = 0$).

From the μ independence of $D^{(0)}$. The same follows from Eq. (4.5). In $d_\perp = 2-\epsilon$, g^2 has mass dimension ϵ, so $\frac{d\alpha_s}{d\log\mu^2} = -\frac{\epsilon}{2}\alpha_s + O(\alpha_s^2)$. Then

$$\frac{\partial}{\partial\log\mu^2}\Big[\frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}\Big] = \frac{\alpha_s}{2\pi}\Big(-\frac{\epsilon}{2}\Big)\Big(-\frac{2}{\epsilon}\Big) = \frac{\alpha_s}{2\pi} \tag{5.3}$$

(`sympy`: ratio 1), and $\frac{dD^{(0)}}{d\log\mu^2} = 0$ gives Eq. (5.2).

Solution to first order.

$$D_{h/g}(\xi,\mu) = D_{h/g}(\xi,\mu_0) + \frac{\alpha_s}{2\pi}\log\frac{\mu^2}{\mu_0^2}\int_\xi^1 \frac{dz}{z} P_{gg}(z)\, D_{h/g}\Big(\frac{\xi}{z},\mu_0\Big) + O(\alpha_s^2). \tag{5.4}$$

With $\mu \sim |\mathbf{k}|$, the logarithms ℓ of Row IV are small, and the large collinear logarithms are resummed by Eq. (5.2).

6 The $\delta(1-\xi)$ part: Row IV and Group I

Row IV alone has no $\frac{11N_c}{6}\delta(1-\xi)$. Its $\delta(1-\xi)$ term is the soft logarithm of Eq. (3.4). The other poles proportional to LO are in Group I, Row II and Row V (part 2b, UV). In `Group_I_corrected.pdf`, with the exact brackets (G5a, G5d), they are:

- **Row II** (per ordering):

$$\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik\cdot(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2(x-w)^2}\,\bar{E}_{x-w}\,B_{\rm II}\,[U(x')-U(w')][U(w)-U(x)]\rho\rho,$$

$$B_{\rm II} = \frac{11}{3} + \log\frac{\Lambda^2}{k^{+2}-\Lambda^2} + \log\Big(1-\frac{k^+}{\vee}\Big) - 8\delta + 2\delta^2 - \frac{4}{3}\delta^3.$$

  The colour factor is $-\mathcal{O}$ with $y\to w$, $y'\to w'$ (numerically: ratio -1). The prefactor relative to LO is $\frac{\alpha_s}{2\pi}\cdot\frac{N_c}{2}$. So, per ordering, Row II $\to -\frac{\alpha_s}{2\pi}\frac{N_c}{2}\bar{E}\,B_{\rm II}\times$LO.

- **Row V, 2b UV** (per ordering):

$$\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\,B_{\rm V}\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\,E_{y-z}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}[U(x')-U(w')][U(y)-U(z)]\rho\rho,$$

$$B_{\rm V} = 2\log\frac{\vee}{\Lambda} - 2\log\frac{\vee}{k^+} + 2\log\Big(1-\frac{\Lambda}{k^+}\Big) - \frac{11}{6} + 4\delta - \delta^2 + \frac{2}{3}\delta^3.$$

  The colour factor is $+\mathcal{O}$ with $x\to y$, $y\to z$, $y'\to w'$ (ratio $+1$). The prefactor is $\frac{\alpha_s}{2\pi}\cdot N_c$. With $E=-\bar{E}$, per ordering Row V 2b $\to -\frac{\alpha_s}{2\pi}N_c\bar{E}\,B_{\rm V}\times$LO.

- **Complex conjugate.** Group I is $\mathcal{C}\cdot\mathcal{V}+$c.c. (Eq. (CS1)), so both rows count twice. Row IV is the square $4\mathfrak{C}'^\dagger\mathfrak{C}'$ and counts once.

- **Charge renormalization.** With $\alpha_0 = \alpha_s(\mu)\big[1-\frac{\alpha_s\beta_0}{4\pi}\frac{1}{\epsilon_{\rm std}}\big] = \alpha_s(\mu)\big[1+\frac{\alpha_s\beta_0}{4\pi}\frac{1}{\bar\epsilon}\big]$, $\mathrm{LO}(\alpha_0) = \mathrm{LO}(\alpha_s(\mu)) + \frac{\alpha_s}{2\pi}\frac{\beta_0}{2}\frac{1}{\bar\epsilon}$LO.

source	weight	coefficient of $\frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}N_c\,\mathrm{LO}(k^+,\mathbf{k})$
Row IV, Eq. (3.4)	1	$2\log\frac{1+\delta}{\delta}$ $\big(=2\delta Y - 2\log\frac{\vee}{k^+} + 2\log(1+\delta)\big)$; and $-R_\Lambda$ (not $\propto$ LO)
Row II	$\times2$	$-B_{\rm II} = -\frac{11}{3} + 2\log\frac{1}{\delta} + \log(1-\delta^2) - \log(1-\xi_0) + 8\delta - 2\delta^2 + \frac{4}{3}\delta^3$
Row V, 2b UV	$\times2$	$-2B_{\rm V} = \frac{11}{3} - 4\log\frac{1}{\delta} - 4\log(1-\delta) - 8\delta + 2\delta^2 - \frac{4}{3}\delta^3$
charge renormalization		$+\frac{11}{6}$
sum		$\frac{11}{6} + 3\log\frac{1+\delta}{1-\delta} - \log(1-\xi_0)$; and $-R_\Lambda$

(`sympy`: sum minus the last line $=0$; here $\log\frac{1}{\delta} = \delta Y - \log\frac{\vee}{k^+}$.)

- **The soft logarithms cancel:** δY appears with $+2$ (Row IV), $+2$ (Row II) and -4 (Row V 2b). This holds only with the weights $1:2:2$. So it also checks the relative normalization of Row IV and of the c.c. in Group I.

- **The $\frac{11}{3}$ cancel** between Row II and Row V 2b (per ordering, as found in `Part2b_MSbar.pdf`), and so do the powers $8\delta - 2\delta^2 + \frac{4}{3}\delta^3$.

- **The $\frac{\beta_0}{2}\delta(1-\xi)$ of P_{gg} is the charge renormalization of LO.** This resolves the open point of `Part2b_MSbar.pdf` (and of the subsection "Renormalization-group bookkeeping" in `sec8_rc_dglap.tex` of the paper draft): no further row is needed to supply $\frac{\alpha_s\beta_0}{4\pi}\frac{1}{\epsilon_{\rm std}}$. With one ϵ for all poles, the pole of the coupling counterterm is not cancelled; it becomes the $\delta(1-\xi)$ term of the fragmentation counterterm (4.4). Together with Row IV it gives the complete P_{gg}, Eq. (4.2).

- **Exact remainder.** For finite Λ/k^+ and $k^+/\vee$, the sum leaves

$$\frac{\alpha_s}{2\pi}\frac{1}{\bar{\epsilon}}\Big\{N_c\Big[3\log\frac{1+\delta}{1-\delta}-\log\Big(1-\frac{k^+}{\vee}\Big)\Big]\frac{d^3N_{\rm LO}}{d^3k}(k^+,\mathbf{k})-R_\Lambda\Big\},\qquad 3\log\frac{1+\delta}{1-\delta}=6\delta+2\delta^3+\dots \tag{6.1}$$

  This is of order $\frac{\Lambda}{k^+}$ and $\frac{k^+}{\vee}$. It is not of the form $P\otimes\mathrm{LO}$, since R_Λ involves $f'(1)$, so it cannot go into $D_{h/g}$.

  - It comes from the different longitudinal limits of the rows: $p^+>\Lambda$ in Row IV; $\Lambda<p^+<k^+-\Lambda$ in Row V 2b; the loop up to $\vee$ in Row II.
  - Either it cancels against rows not included here, or the limits must be made consistent.
  - This is still open, as was already noted in `Part2b_MSbar.pdf` for the poles multiplying $\log(1-\frac{k^+}{\vee})$ and the powers of $\frac{\Lambda}{k^+}$.

7 Checks

Script `frag_checks.py`, results in `frag_checks.json`.

check	result
your Eq. (1.1) minus Eq. (1.5) of `RowIV_dperp.pdf` (`sympy`)	0
conversion factor $\frac{\alpha_s}{2\pi}\frac{N_c}{\bar{\epsilon}^3}$ (`sympy`)	0
convolution form (2.6) minus compact form (`sympy`)	0
$d_\perp=2-\epsilon$: $\frac{d_\perp\bar{E}-2}{2}-\big(\frac{1}{\bar{\epsilon}}-\ell\big)$ at $O(\epsilon^0)$ (`sympy`)	0
plus prescription with exact limits (3.2), 24 random cases	rel. 1×10^{-13}
$R_\Lambda/\delta\to-2N_c[f(1)+f'(1)]$	-1.97000, -1.97021, -1.97025 vs. -1.97025
momentum sum rule, $\delta(1-\xi)$ coefficient (`sympy`)	$\frac{11N_c}{6}\delta(1-\xi)$
Jacobian identity (4.6)	rel. 4×10^{-14}
$\frac{\partial}{\partial\log\mu^2}\big[\frac{\alpha_s}{2\pi}\frac{1}{\bar{\epsilon}}\big]\big/\frac{\alpha_s}{2\pi}$ (`sympy`)	1
colour factors of Row II and Row V 2b relative to $\mathcal{O}$	-1, $+1$
$\delta(1-\xi)$ tally, exact in δ and ξ_0 (`sympy`)	0